

TaichiDough: Reconstructing and Simulating Deformable Dough

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Abstract

TaichiDough is a research prototype for reconstructing, simulating, and evaluating deformable dough from recorded point clouds and the motion of two tools. It combines a Taichi MLS-MPM simulator with a calibration and evaluation workflow for comparing simulated deformation with recorded observations in metric scene coordinates. The current repository demonstrates a good legacy static initialization match and timestamped replay, while its longer proxy-tool rollout identifies the remaining work required for physically validated prediction.

1 From Recorded Observation to Simulation

The workflow begins with recorded DeformPath point clouds and captured trajectories for two tools. A scene calibration maps camera measurements into simulation coordinates. The intended metric procedure uses an AprilTag, exact camera intrinsics, a measured `scene_from_tag` transform, and a floor plane. The calibration collector checks the tag family and ID, tag edge length, timestamps, frame identifiers, and the TF path between the point-cloud and colour-camera frames. It rejects pose outliers before writing a version-2 record with transforms, intrinsics, diagnostics, and input fingerprints.

Calibration Algorithm

AprilTag detection estimates a rigid pose $T_{C \leftarrow A}$ from the tag frame A to the camera frame C using the tag’s known physical edge length and camera intrinsics. The collector accepts multiple valid detections of the fixed tag. Each pose is represented as

$$T = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix},$$

where $R \in SO(3)$ is rotation and $t \in \mathbb{R}^3$ is translation. It first takes the coordinate-wise median of observed translations, then measures translation residuals as Euclidean distances from that median. Rotation residuals are geodesic angles on $SO(3)$, measured against the rotation with the smallest median pairwise angular distance.

For each residual type, the acceptance threshold is the smaller of a fixed safety limit and a robust median-absolute-deviation threshold:

$$\tau = \min(\tau_{\max}, \text{median}(r) + 3.5 \times 1.4826 \times \text{MAD}(r)).$$

Detections must pass both tests. Accepted translations are averaged arithmetically. Accepted rotations are averaged with the Markley quaternion method: normalized quaternions q_i form $A = \sum_i q_i q_i^T$, and the unit eigenvector of A with the largest eigenvalue supplies the mean rotation. A final fixed residual check removes any remaining outlier before the pose is recomputed.

Given a measured scene-to-tag transform $T_{S \leftarrow A}$, the metric camera transform is

$$T_{S \leftarrow C} = T_{S \leftarrow A} T_{C \leftarrow A}^{-1}.$$

If the point cloud has source frame P , TF provides $T_{C \leftarrow P}$ and the point cloud is mapped with $T_{S \leftarrow P} = T_{S \leftarrow C} T_{C \leftarrow P}$. This is a rigid calibration: it does not fit an axis permutation, scale, or post-hoc registration.

The checked-in Episode 18 benchmark instead uses a legacy bootstrap calibration: an axis remapping, uniform scale, translation, and prescribed top-view camera. This is useful for reproducing existing benchmarks, but it is not an independently verified metric calibration. The repository contains the AprilTag procedure needed to replace it; no completed version-2 calibration result is committed.

After calibration, the reconstruction program converts the observed visible dough into a particle initialization. In floor-fill mode, it removes points at or below the calibrated floor clearance, groups retained points by occupied columns, and fills each column down to the floor. The voxel count gives the reconstructed volume. A deterministic particle sample is accompanied by metadata recording the calibration, floor, volume, particle count, and file fingerprints. With measured dough mass, particle volume, density, and mass are derived from the reconstructed volume and total mass, preserving those quantities when numerical resolution changes.

2 MPM Simulator

The simulator uses three-dimensional Moving Least Squares Material Point Method (MLS-MPM) with Affine Particle-In-Cell (APIC) transfer. Each material particle stores position, velocity, deformation gradient, and affine velocity; optional plastic-volume state is also supported. Momentum and mass transfer between particles and a regular background grid at every step, while the grid integrates gravity and collision response.

The constitutive law combines corotated elastic stress with a simple viscous term. Young’s modulus and Poisson ratio determine the elastic Lamé parameters. Plastic deformation can be represented by clamping the singular values of the deformation gradient, optionally with plastic-volume hardening and damping. This is a simplified dough-like model, rather than a completed rheological model of food dough.

Core Simulation Equations

For particle deformation gradient F_p , APIC affine velocity C_p , and timestep Δt , the deformation update is

$$F_p^{n+1} = (I + \Delta t C_p) F_p^n.$$

The Lamé parameters are computed from Young’s modulus E and Poisson ratio ν :

$$\mu = \frac{E}{2(1 + \nu)}, \quad \lambda = \frac{E\nu}{(1 + \nu)(1 - 2\nu)}.$$

With polar decomposition $F_p = RS$, determinant $J = \det(F_p)$, and viscosity η , the stress-like term is

$$P = \underbrace{2\mu(F_p - R)F_p^T + \lambda J(J - 1)I}_{\text{corotated elastic contribution}} + \underbrace{\eta(C_p + C_p^T)}_{\text{viscous contribution}}.$$

For quadratic B-spline weight w_{pi} between particle p and grid node i , particle mass m_p , particle volume V_p , grid spacing Δx , and particle-to-node offset d_{pi} , the particle-to-grid update is

$$m_i += w_{pi}m_p, \quad m_i v_i += w_{pi} [m_p v_p + (-\Delta t V_p 4\Delta x^{-2} P + m_p C_p) d_{pi}].$$

Node velocity is normalized by accumulated mass and updated with gravity, then interpolated back to particles to update v_p and C_p . If plasticity is enabled, $F_p = U\Sigma V^T$ is decomposed by singular value decomposition and each singular value is clamped to the configured interval $[\sigma_{\min}, \sigma_{\max}]$ before rebuilding F_p .

For tool contact, the velocity at particle position x is

$$v_{\text{tool}}(x) = v_{\text{linear}} + \omega \times (x - c),$$

where c is the tool centre and ω is angular velocity. The simulator forms relative velocity $u = v_p - v_{\text{tool}}(x)$, removes an inward normal component when $u \cdot n < 0$, then applies the configured friction, absorption, and stickiness. A post-advection projection also moves particles out of the floor or tool volume when penetration remains.

A constant-height floor and two moving tools provide contact. Recorded replay can use calibrated oriented boxes or signed-distance fields generated from the supplied UR and Kinova spatula meshes. The contact calculation accounts for both linear and angular tool velocity. Recorded positions are interpolated linearly and orientations use shortest-path spherical interpolation. The simulator also produces rendered depth maps, virtual point clouds, video, GUI output, UDP observations, and interfaces for Gymnasium and Stable-Baselines3 experiments.

3 Evaluation and Calibration Attempts

The static Episode 18 benchmark evaluates the reconstructed initial state under the bootstrap calibration. It contains 10,463 filtered observed points and compares visible simulated and recorded geometry without translating, scaling, or registering either point set. The footprint intersection-over-union is **0.8808**. The simulated-to-reference extent ratios are **1.058** and **1.052**, the area ratio is **1.113**, centroid offsets are below **0.5 mm** on both top-view axes, and the 95th-percentile depth error is **2.0 mm**. These results support the static visible-shape initialization, but they do not establish material-model accuracy.

The longer dynamic replay gives a more demanding result. It replays captured trajectories over 66 timestamp-paired observations using two identical 5 cm half-extent proxy boxes, a 3,000-particle simulation, a 24^3 grid, a 0.2 ms timestep, and the default Young’s modulus of 2,000 Pa. Across the 65 post-initial frames, mean pixel intersection-over-union is **0.4240**, mean depth mean-absolute error is **95.95 mm**, mean 95th-percentile depth error is **119.40 mm**, and mean depth-change error is **83.83 mm**.

The dynamic result explicitly records that physical fidelity is not validated. Its known limitations include unmeasured proxy tool geometry, an unverified camera calibration, no tool occlusion in the depth export, assumed hidden volume and material parameters, and visible-geometry-only evaluation. It is therefore a replay and diagnostic result, not evidence of a calibrated physical prediction. A short implementation-smoke replay has substantially better early-frame agreement, but its limited duration only demonstrates that the replay path operates correctly.

4 Material Identification and Next Steps

TaichiDough includes a reproducible procedure for identifying an effective Young’s modulus. Each candidate begins from the same reconstructed particles, replays the same tool trajectory, and varies only Young’s modulus. Candidate runs are compared using visible depth change, mask overlap, observed-to-simulated distance, and visible coverage. The search uses a logarithmic coarse grid, local refinement, boundary expansion, cached complete evaluations, and bootstrap intervals over deformation windows. Training windows select candidates, while held-out windows assess prediction without refitting.

No completed material-calibration output is included in this checkout. When obtained, the selected value must be reported as an effective Young’s modulus for the selected constitutive model, viscosity, plasticity, contact method, reconstructed volume, mass assumption, and numerical resolution; it is not a universal constant for dough. The next experimental priorities are to collect a verified AprilTag metric calibration, measure tool geometry and marker transforms, use measured dough mass, run held-out material calibration, and repeat the fit at several particle counts, grid resolutions, and timesteps.

5 Conclusion

TaichiDough provides an end-to-end basis for data-conditioned dough simulation: calibration, floor-aware reconstruction, mass-preserving MPM initialization, timestamped two-tool replay, visible-geometry comparison, and material-parameter search. The static benchmark and replay implementation are established. The dynamic proxy result also makes the current limitations measurable and identifies the measurements needed before claiming physical agreement.